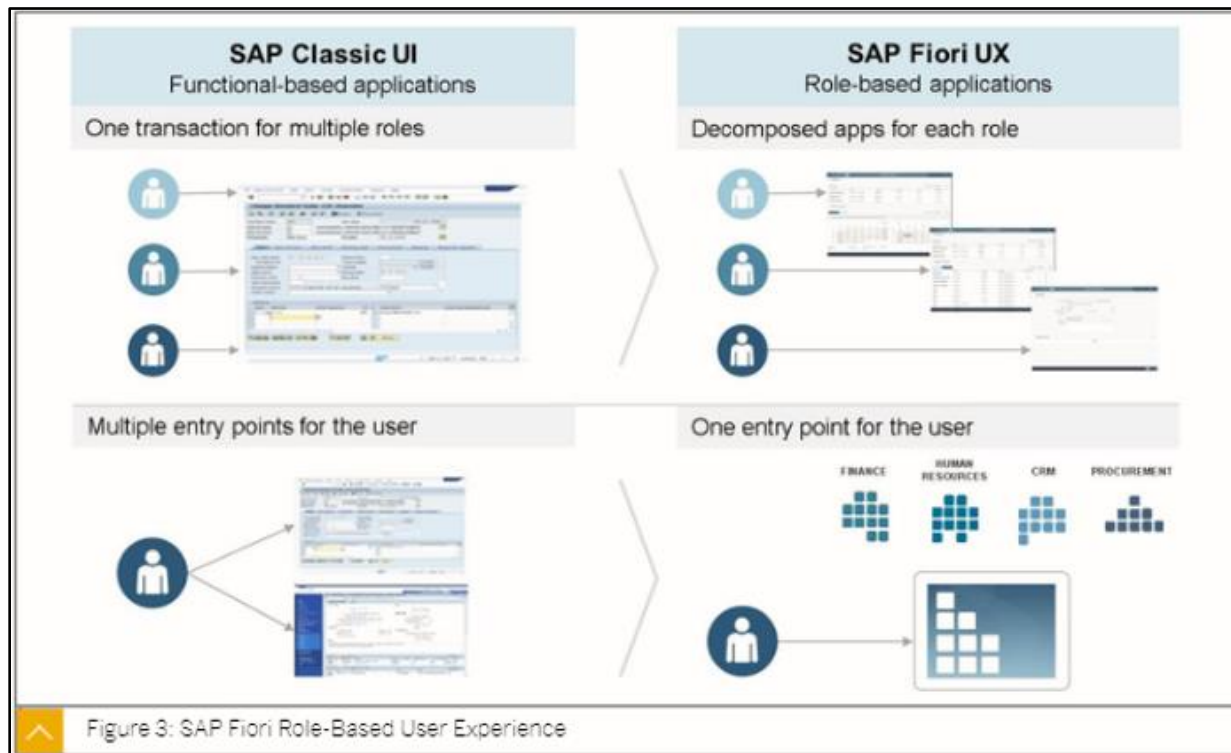


CONTENTS

1. SAP FIORI CONCEPT
2. SAP FIORI ARCHITECTURE
3. Types of Fiori Applications
4. Configuration of Fiori Applications
5. Roles and Authorization Concept
6. Troubleshooting of Fiori Apps
7. Business Requirements from Fiori
8. Some Important T-code and Table

SAP FIORI CONCEPT

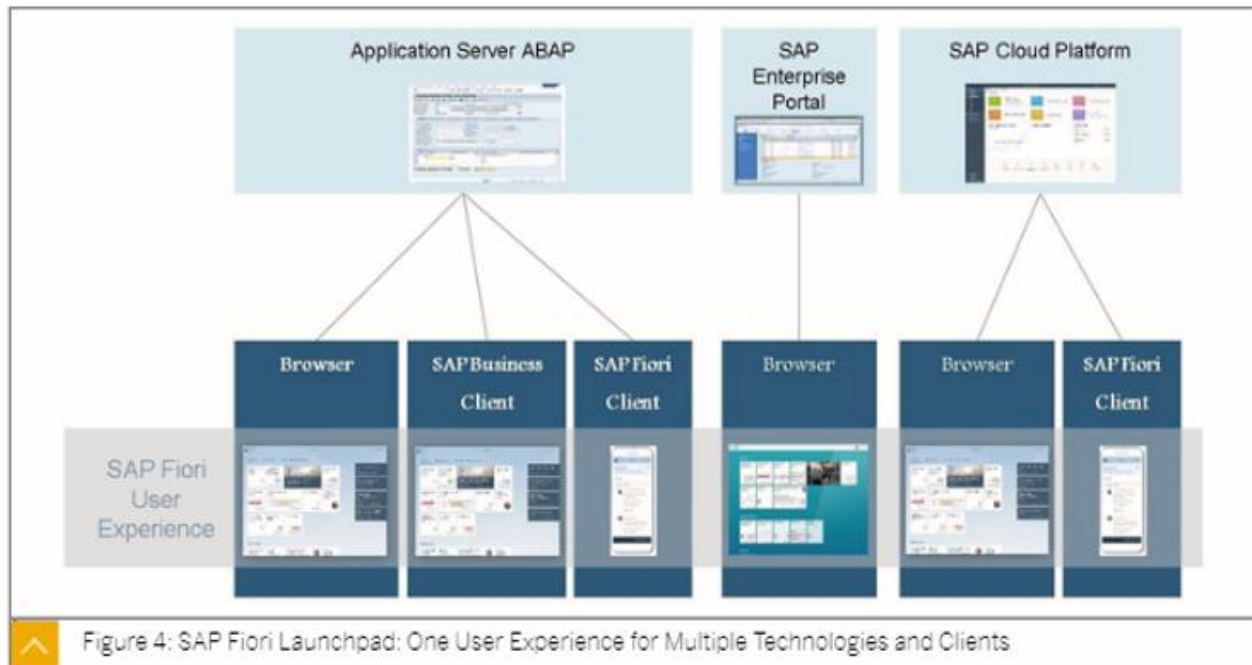
From the below picture you can have an overview of what is the concept of Fiori.



As you can see the classic SAP user interface (UI) offers a single complex transaction for many user roles the task of a single user might then require multiple transaction or additional application.

SAP Fiori breaks down these big transactions into several discrete apps, suiting the user roles, all apps are connected to each other, so that all the tasks of the transaction are possible but it is not necessary that the user will have access to all the apps, that is depending on the user to role mapping.

SAP Fiori Launchpad then serves as a central entry point (Can be set as a single entry point) for all the apps of a user, so that the user doesn't have to log on to different clients and platforms. See below.

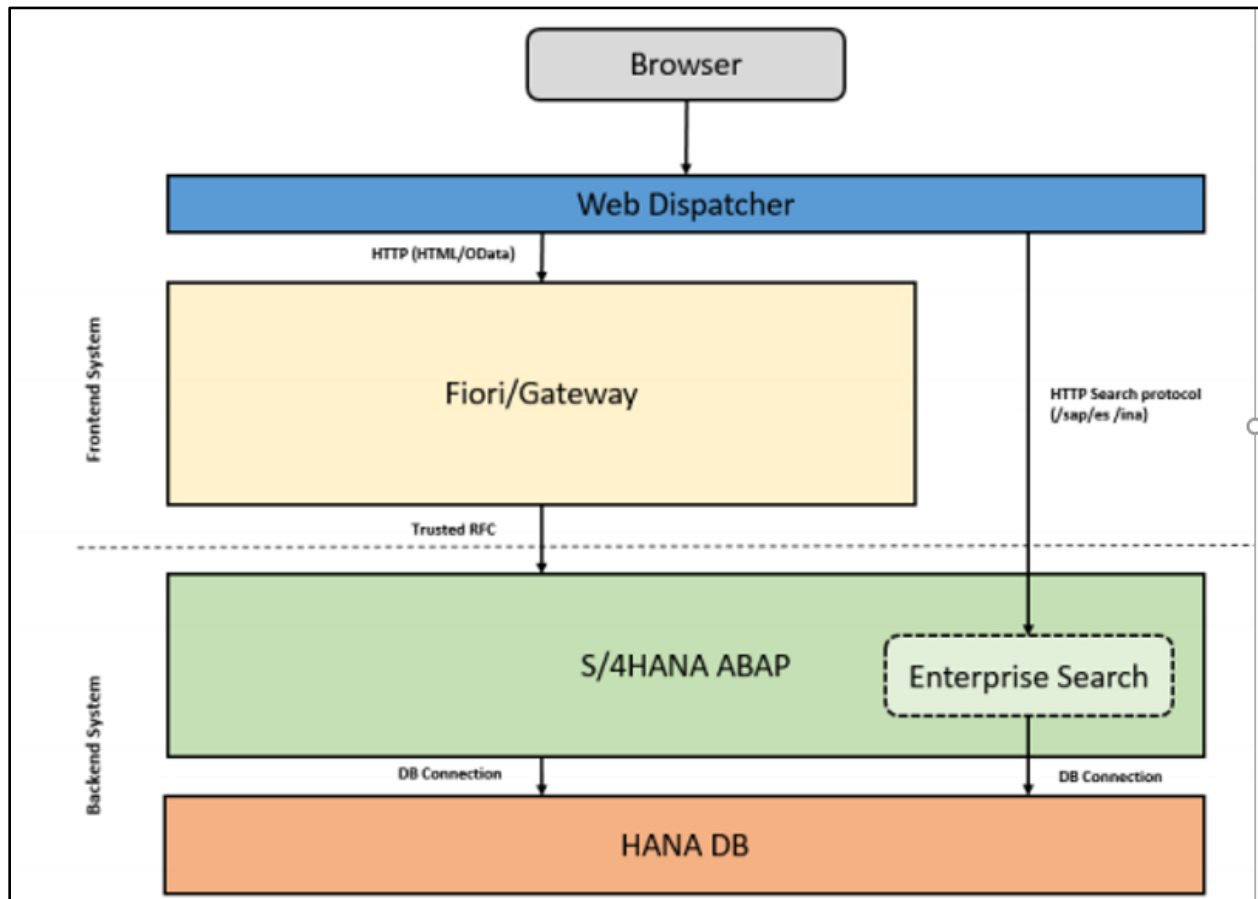


SAP Fiori Launchpad can be run on SAP NetWeaver ABAP Stack (ECC/S4HANA), SAP Enterprise Portal, SAP business client, SAP Fiori Client & SAP cloud platform. Most commonly in all the program SAP Fiori Launchpad run on SAP NetWeaver ABAP Stack.

SAP Fiori Launchpad is not technically the same on other platform therefore some feature of SAP Fiori Launchpad on SAP Netweaver ABAP Stack may not be available on other platform, Also the configuration and authorization is different from platform to platform, However the concept of the user's experience will be the same.

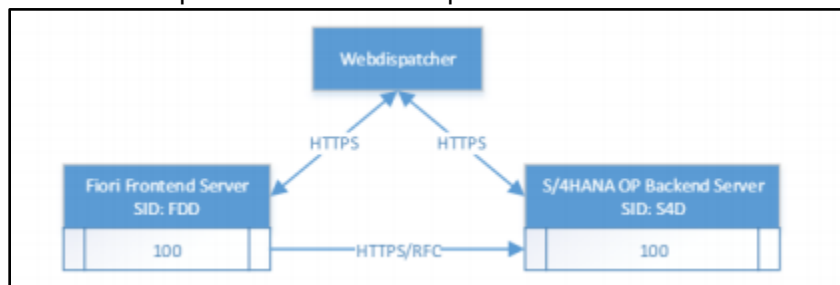
- Demonstrate the Launch Pad of Fiori and their functionality.

SAP FIORI ARCHITECTURE



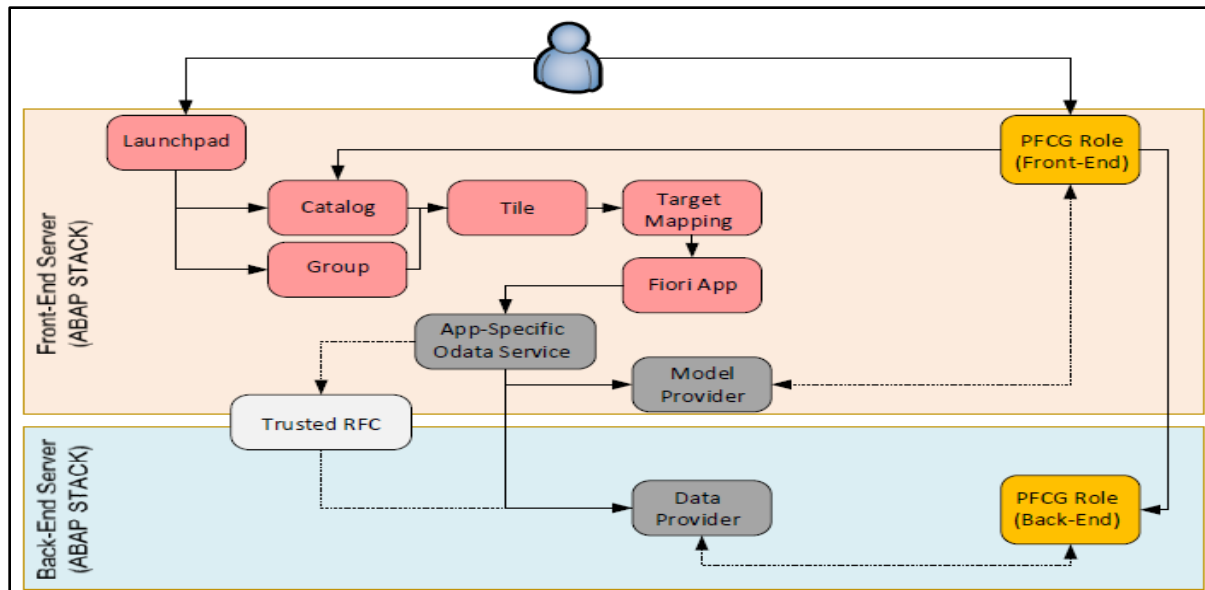
Web Dispatcher will receive all the Http(s) request and direct to Fiori Gateway System.

Below is the picture how web dispatcher connects with S4HANA and Fiori System.



Above 2 image is just for understanding the structure but moreover that is related to basis activity.

As a security team we need to understand the below structure.



As we can see the diagram is separated in 2 parts

1. Front End (Fiori)
2. Backed (S4 HANA)

Note: In case of embedded system both Front End (Fiori) and Back End (S4 HANA) will be in same box. We will discuss the difference, Pros & Cons of Embedded system and Separately.

Component of Front End (Fiori).

Catalog is a set of apps you want to make available for one role. Depending on the role and the catalog assigned to the role, the user can browse through the catalog, choose apps from this catalog, and add them to the entry page of the SAP Fiori Launchpad.

Group is a subset of catalog that contains the apps visible on the SAP Fiori Launchpad entry page. Which tiles are displayed on a user's entry page depends on the group assigned to the user's role.

Tiles can offer summarized information about the data available for the users in the app. Depending on the roles of the user certain groups of apps are displayed.

Apps Each app is represented by a tile, App is the combination of tile configuration and target mapping.

Target Mapping which define the target application which is launched when an intent (semantic object/action) is triggered

Launchpad We can call it as a Fiori Launch Pad (Home page for the end users) where in all the Apps are available in the form of groups for the users based on the roles assignment.

O-Data

- SAP Fiori apps use OData to display and update data in back-end server (SAP or Non-SAP).

- OData (Open Data Protocol) is a standardized protocol for creating and consuming data APIs.
- OData builds on core protocols like HTTP and commonly accepted methodologies like REST.

RFC Connection – RFC connection should be trusted to connect front end & Backend.

Any Gateway service is implemented by 2 ABAP classes - **Model Provider** class & **Data Provided Class**.

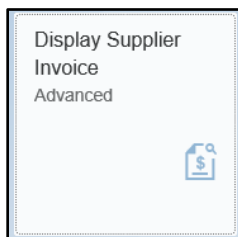
Roles need to exist in both front-end and back-end systems

- SAP Fiori (Front-End) – controls what tiles user can see
- S/4 HANA (Back-End) – Controls what actions a user can perform.

Note: In case of Embedded System Only one role can be acquire to have authorizations and catalog, groups.

Fiori Application Types

1. **SAP GUI Apps** – These are the T-code based apps and require t-code access on the backend.
Example,



Display supplier invoice is a GUI app because it requires **s** T-code access at the backend.

Some facts of SAP GUI app:

- SAP GUI & Web Dynpro apps opens in a new tab in a browser unlike native Fiori apps.
- APP ID in Fiori Library will be same as T-code, just as above case The app ID for Display Supplier Invoice is **MIR4** you can see the below details in Fiori Library.

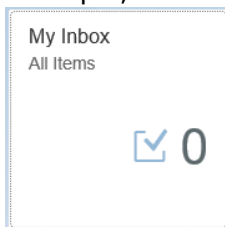
Call MIRO - Change Status, Display Supplier Invoice - Advanced

SAP S/4HANA
Required Back-End Product SAP S/4HANA
Application Type SAP GUI
Database HANA DB exclusive
Form Factor Desktop
App ID MIR4

2. **Transactional (SAP UI5)** – These are native Fiori apps and require access to O-data services on the backend.

Transactional app also called task based app - Access to task like change, create or entire process with guided navigation as well as reused component of shared features.

Example,



My Inbox is a transactional app and the app ID is **F0401A** unlikely as GUI app.

Transactional app sounds like a T-code based app but it is not related to any T-code.

Transactional app work based on the O-data unlikely as GUI app.

Transactional Apps opens in the same tab like GUI app opens in the new tab on browser.

You can see the details in Fiori Library.

My Inbox - Approve Purchase Requisitions

for Manager - Procurement

SAP S/4HANA
Required Back-End Product SAP S/4HANA
Line of Business Sourcing and Procurement, Supply Chain
Application Type Transactional (SAP Fiori: My Inbox)
Database HANA DB exclusive
Form Factor Desktop, Phone, Tablet
App ID F0401A

3. **Analytical/Design-studio** (Insight to Action) – These are basically reports and require access to either O-data or BEX query on the backend. For analytical apps which have design studio as their application type, we require to provide access to BW reports/CDS views on the backend.

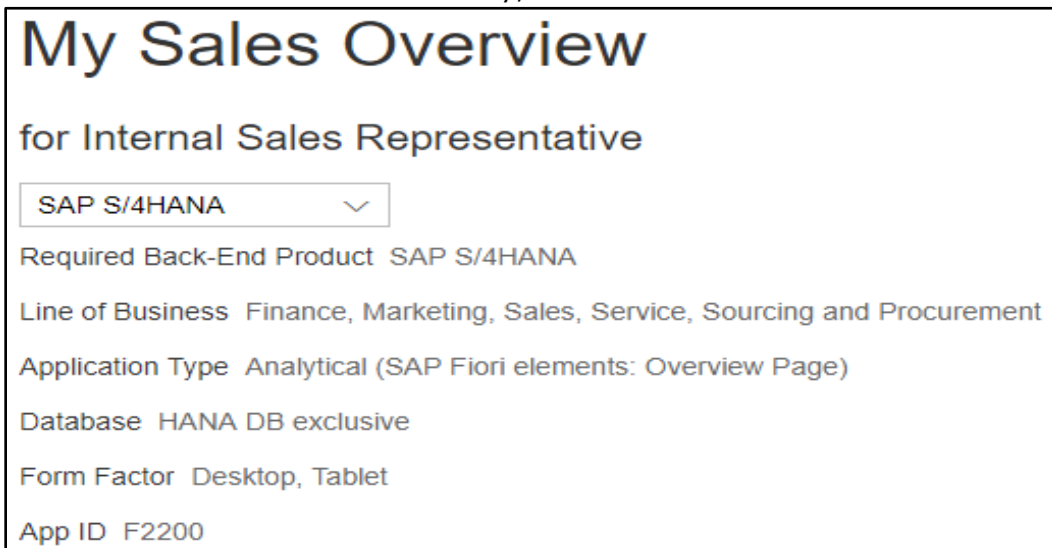
Analytical app gives visual overviews over a complex topic for monitoring or tracking purposes.

Example,



My sales overview is an Analytical app which shows the reporting view in real time. Analytical app is only for reporting purpose.

You can see the details in Fiori Library,

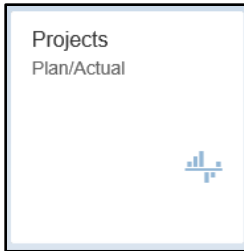


4. **Web-Dynpro Apps** – These are Web Dynpro type apps which require Web Dynpro Application access on the backend.

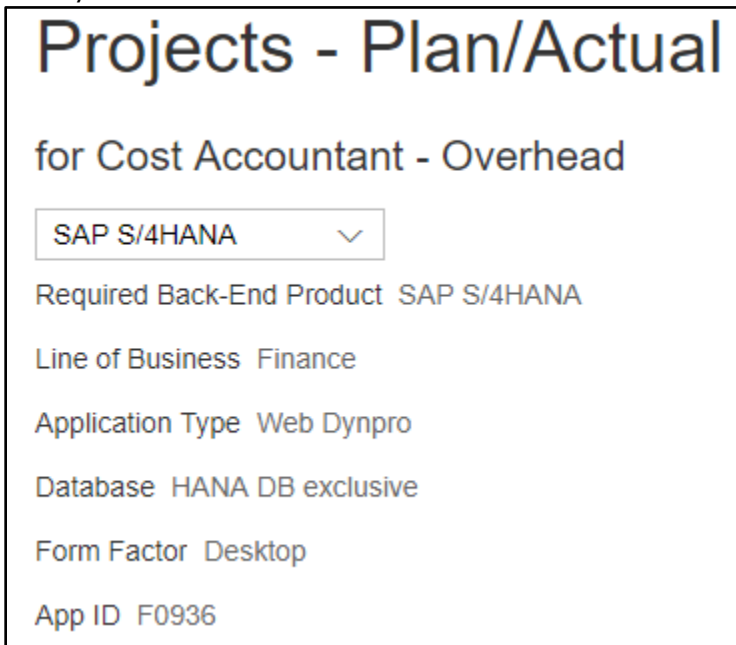
Fiori Launchpad uses the nwbc service to launch the Web Dynpro applications. Which means although the Web Dynpro application is working fine on your backend server, if you want to integrate it to Fiori Launchpad you need to activate the following ICF node: **/sap/bc/ui2/nwbc**

WebDynpro and GUI Apps are Legacy apps for HTML application they are not mobile enabled but provide a SAP Fiori look and feel.

Example,



Project Plan/Actual is a webDynpro app you can see the details below from fiori library.



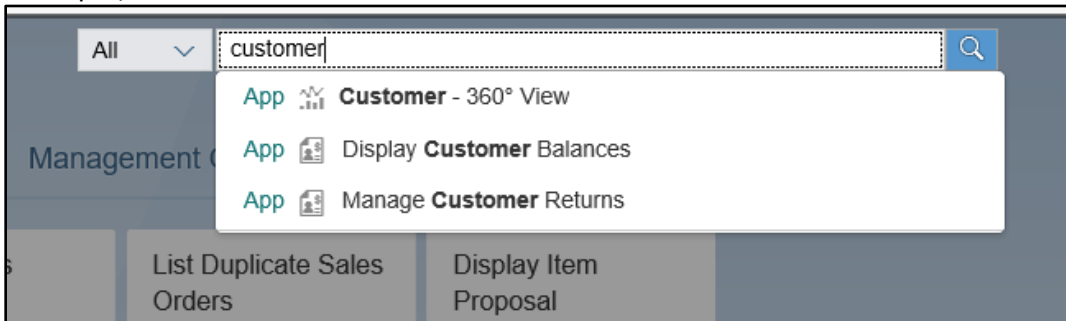
WebDynpro apps also opens in a new tab in the browser.

Both GUI & Webdynpro app is controlled by system alias for navigating to backend system.

5. **FactSheet Apps** (Search and Explore) - Fact Sheet apps are not tile based apps rather Fiori search based. Fact Sheets are called by search results from Fiori Launchpad. As a pre-requisite, Enterprise Search need to be activated in backend system and search requests from Fiori Launchpad hits the backend system directly.

FactSheet app view on essential information about an object and contextual navigation between related object.

Example,



Fact Sheet are different from all other apps as you can see there is no separate tile for Customer 360-degree view app unlikely other apps as fact Sheet app only have the target mapping details during configuration of this app in Fiori Library

You need to search the app name in the enterprise search bar.
You can get the details from Fiori Library.

Customer - 360° View

for Internal Sales Representative

SAP S/4HANA

▼

Required Back-End Product SAP S/4HANA

Line of Business Finance, Marketing, Sales, Service, Sourcing and Procurement

Application Type Fact sheet (SAP Fiori elements)

Database HANA DB exclusive

Form Factor Desktop, Phone, Tablet

App ID F2187

Configuration of Fiori Tiles

Teams Involved

Basis – For any native fiori app configuration, we require activation of certain services which is done by Basis team. These services include O-data services and the SICF nodes.

Security/Development – Once the activation of these services is accomplished, Security/Development team can start the configuration of the tiles.

Note: Native Fiori apps will not work without the activation of these services.

Important T-codes

/n/ui2/flpd_cust – to invoke Fiori Launchpad Designer (for configuration of Fiori tiles)

/n/ui2/flp – to invoke Fiori Launchpad (for end user access)

/n/iwfnd/maint_service – To activate/maintain the O-data services

Before we start configuring the tiles let's see all the parameters/Keywords of Fiori Tiles & Target Mapping.

Taking an example of App ID **MIR4** i.e Display Supplier Invoice

Below is the standard tile configuration of app ID **MIR4**

The screenshot shows the configuration interface for a Fiori tile. The title bar indicates 'Configure: Display Supplier Invoice' and 'Instance ID: FA163EDF73161ED59DE8784EC3A6E0D2_AL'. The interface is divided into two main sections: 'General' and 'Navigation'.

General Section:

- Title: Display Supplier Invoice
- Subtitle: Advanced
- Keywords: (empty field)
- Icon: sap-icon://Fiori2/F0390
- Information: (empty field)

Navigation Section:

- Use semantic object navigation: ☒
- Semantic Object: SupplierInvoice
- Action: displayAdvanced
- Parameters: sap-ui-tech-hint=GUI
- Target URL: #SupplierInvoice-displayAdvanced?sap-ui-tech-hint=GUI

Tile Actions Table:

☐	Menu Item	Target T...	Navigation Target	Action	Icon
☐		URL			

At the bottom right, there are 'Save' and 'Cancel' buttons.

Some Keywords of Tiles configuration,

Tile – Name of the Tile

Subtitle – Name of the subtitle of the tile, this is not mandatory field but if you mention anything in Subtitle just as above example, it will appear in the 2nd line of the tile.

Semantic Object - Semantic Object: Represents a business entity such as a customer, a sales order, or a product. Using semantic objects, you can bundle applications that reflect a specific scenario. ... You can either use semantic objects shipped by SAP, or create new semantic objects

Action – Represent the Action/Actvt of any app like, Create, change, Display, Delete.

All above keywords are just to represent the tile/App based on their functionality & Action but it is actually controlled by backed object authorization only.

Below is the standard target mapping configuration of App ID **MIR4**

Configure: 'Display Supplier Invoice - Advanced' Instance ID: FA163EDF73161ED59DE8784EC3A6E0D2_TM

Intent

Semantic Object: SupplierInvoice

Action: displayAdvanced

Target

Application Type: Transaction

Title: Display Supplier Invoice - Advanced

Transaction: MIR4

System Alias: S4PRC

General

Information:

Device Types: ☒ Desktop ☐ Tablet ☐ Phone

Parameters:

☐	Name	Mandatory	Value	Is Regular Expression	Default Value	Target N...
☐	SupplierInvc	☐		☐		RBKP-BELN
☐	FiscalYear	☐		☐		RBKP-GJAF
☐						

Save Cancel

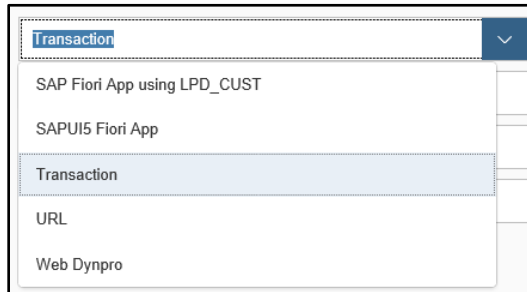
Target mapping is the most important thing in app configuration.

Target mapping is part of the SAP Fiori launchpad configuration. It defines the target application, which is launched when clicking on a tile, on a link or within an app-to-app navigation

Semantic Object, Action & Tile name should be same for any respective tile Just an example of MIR4 App - Title, Semantic Object & Action are same in Tile configuration and Target Mapping Configuration.

Important fields in Target Mapping Configuration apart from Title, Semantic Object & Action.

Application Type – For all the different type of app we need to select the different option from drop down selection.



A screenshot of a dropdown menu in a software interface. The dropdown is open, showing a list of application types. The top item is 'Transaction', which is highlighted in blue. Below it are 'SAP Fiori App using LPD_CUST', 'SAPUI5 Fiori App', 'URL', and 'Web Dynpro'. The dropdown is enclosed in a light gray border with a small blue arrow icon on the right side of the header.

Application Type
Transaction
SAP Fiori App using LPD_CUST
SAPUI5 Fiori App
URL
Web Dynpro

In our case as MIR4 is a GUI app and It require Transaction code access at the backend we will select application type as a **Transaction**.

Transaction -- We need to enter the **T-code** with respect to the GUI type app.

In above case we have,



A screenshot of a form field. The field is labeled 'Transaction:' and contains the text 'MIR4'. The field is enclosed in a light gray border.

Transaction:	MIR4
--------------	------

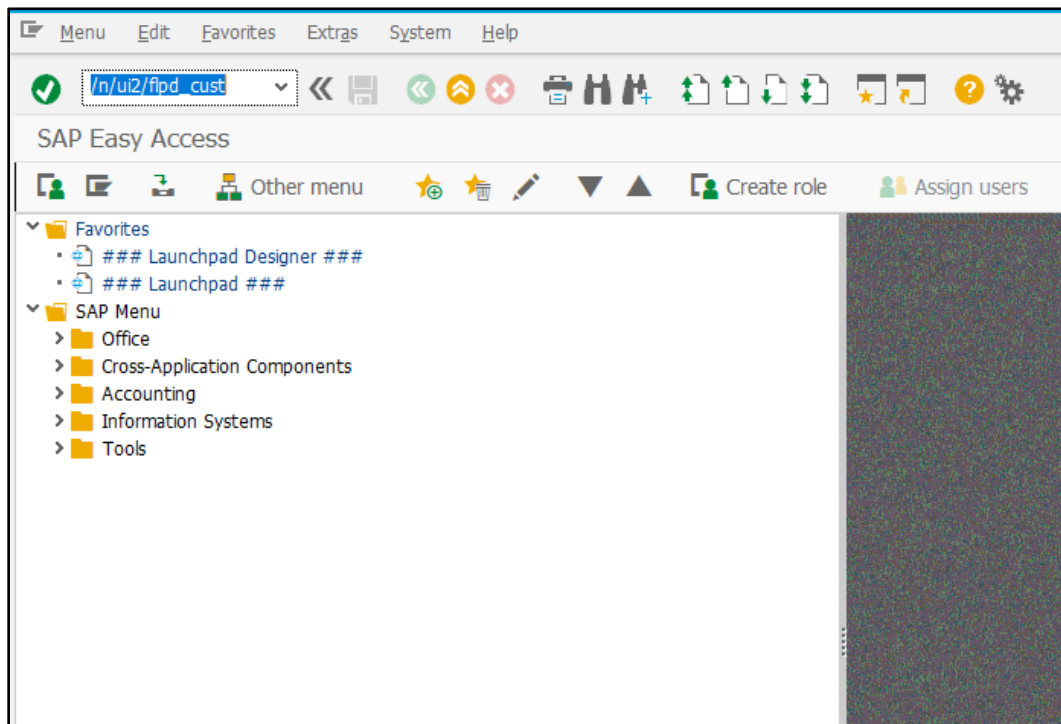
System Alias: The system alias is the result of the routing for an inbound request on SAP Gateway. For the SAP Fiori system landscape, you need one system alias pointing to the back-end server with the indicator

Note: In case of embedded system the Alias name should be **LOCAL** as the app should be pointing to the same system.

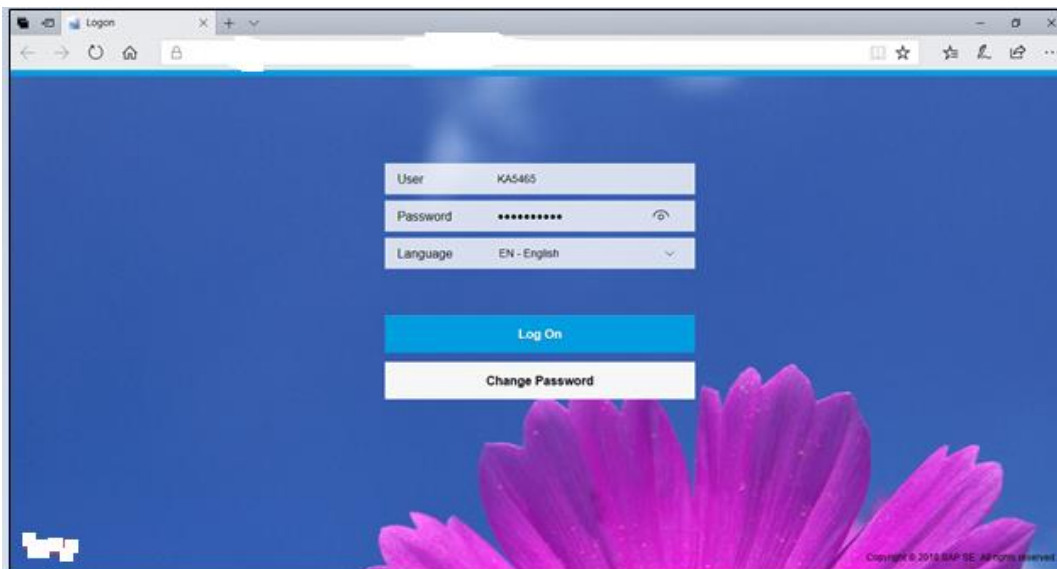
Tile Configuration Process (Front End)

The front end configuration process is the same for any Fiori app. Front end configuration will be done in the Fiori Launchpad Designer. Fiori Launchpad Designer is invoked from the Gateway Fiori client.

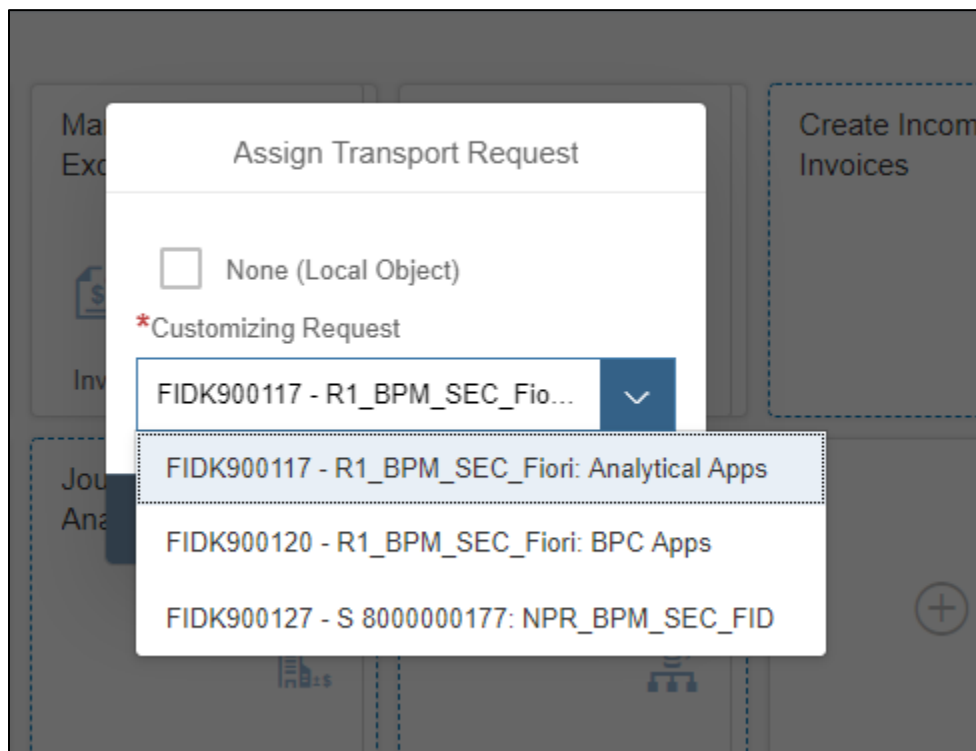
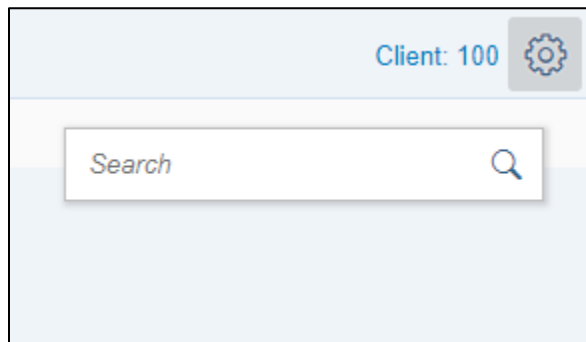
- i. Login to Fiori system. Execute the T-code `/n/ui2/flpd_cust` to invoke the Fiori Launchpad Designer.



- ii. Login to the Fiori Launchpad Designer



Before starting the configuration, select the Transport Request to capture the required changes. Go to the Settings icon on the top right corner of the screen and choose the suitable TR.



After selecting the Transport (Customizing request)

Step 1 is to create the catalog & Groups

Click on + Icon from the current page shown by arrow sign.

The screenshot shows the SAP Fiori Catalogs and Groups interface. The left sidebar contains a 'Catalog Collection' with a search bar and a list of catalogs. The main area displays a grid of tiles. A blue arrow points to a '+' icon at the bottom left of the main area, indicating the option to create a new catalog or group.

Create Catalog

☒ Standard ☐ Remote

*Title:

Analytical Tiles

*ID:

ZFMC_Analytical

Save Cancel

Create Group

*Title:

FMC Analytical Tiles

*ID:

FMC_Analytical

Group personalization:

☐ Enable users to personalize their group

Save Cancel

Start configuring the tiles into the Z catalog. Now, let's say if we want to configure the "Foreign Exchange Overview", we first get the following details from the Fiori apps Library.

1. SAP Business Catalog
2. SICF Node
3. O-data Service/T-code/WDA/BEX Query depending on the App Type.
4. Target Mapping

Note: All the end to end information of apps is present in Fiori Library for any requirement as a part of security team we need to take the reference from Fiori Library.

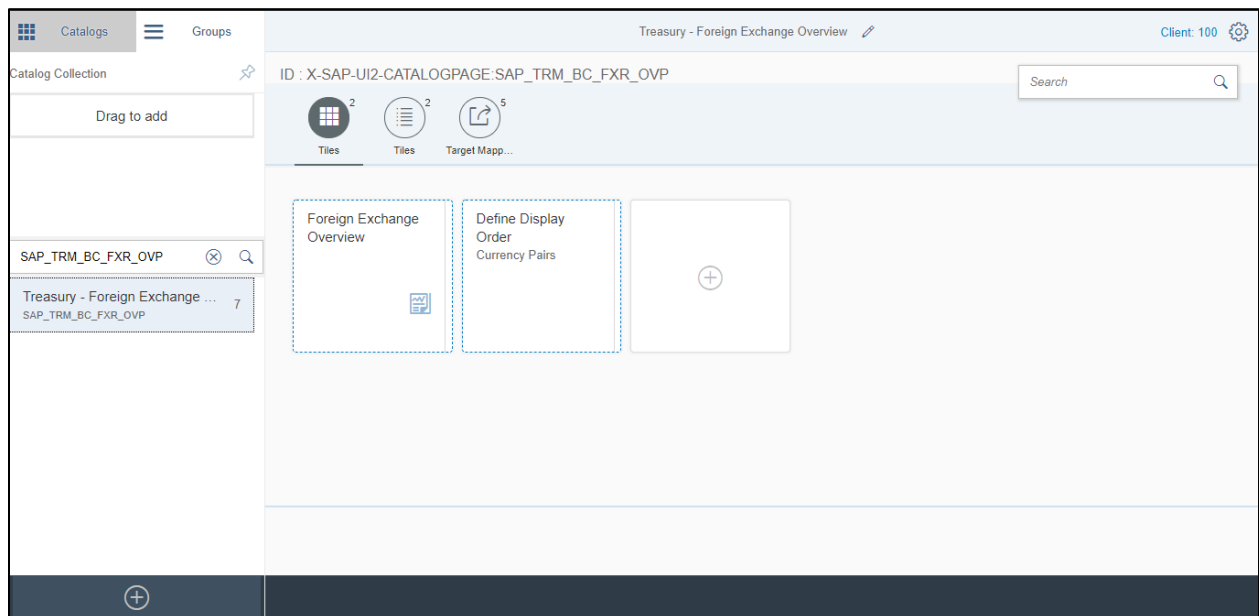
Fiori Library is an open source any one can access it, below is the link to access Fiori Library.

<https://fioriappslibrary.hana.ondemand.com/sap/fix/externalViewer/#/home>

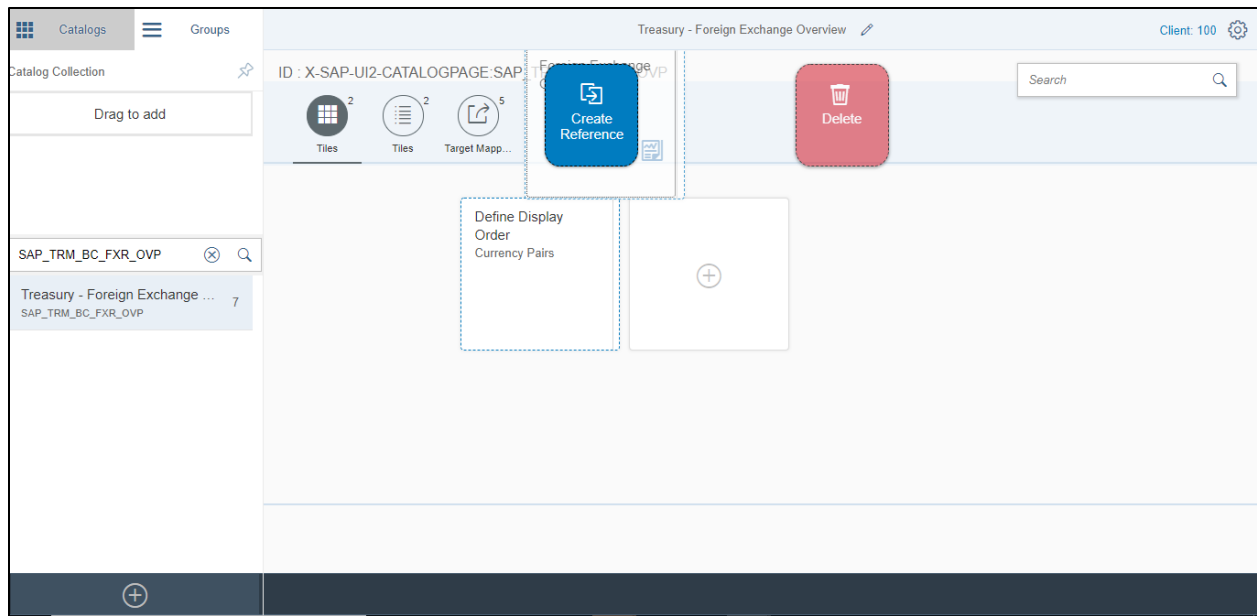
You can search any apps which is published by SAP till date will be available.

For the app Foreign Exchange Overview, Business catalog is SAP_TRM_BC_FXR_OVP (from Fiori Library)

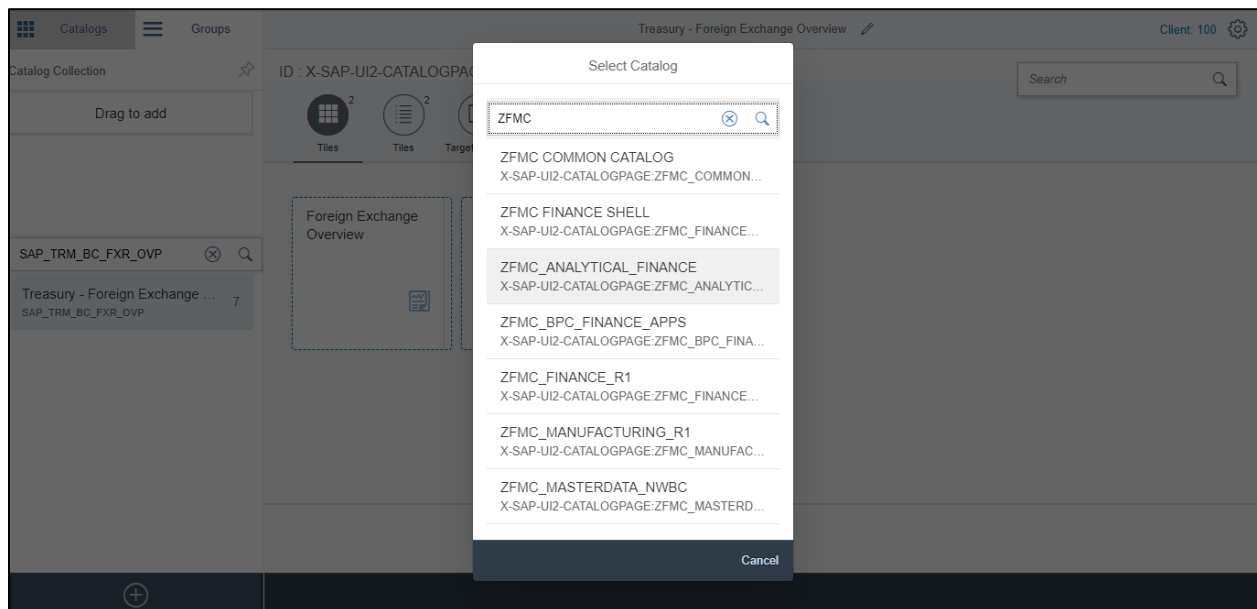
Search for the Business catalog in the Launchpad Designer. From this catalog, we need to reference the tile to our Z catalog.



Drag the tile to the top and add it to the Create Reference tab.



Select the catalog into which you want to add the tile i.e the Z Catalog



Now go to the Target Mapping option next to tiles option. Select the target mapping for the tile "Foreign Exchange Overview" and again Create Reference for the same.

http://usgbfed00.usgboral.sap.local:8000/sap/bc/ui5/ui5/sap/arsvc_upb_admn/main.html?scope=CUST&sap-client=100&sap-language=EN#/Cat

usgbfed00.usgboral.sap.local

Catalogs Groups

Catalog Collection

Drag to add

ID : X-SAP-UI2-CATALOGPAGE:SAP_TRM_BC_FXR_OVP

Search

Tiles Tiles Target Map...

SAP_TRM_BC_FXR_OVP

Treasury - Foreign Exchange Overview

Semantic Object	Action	Navigation Type	Information	Desktop	Tablet
ForeignExchangeOverview	analyzeFXRates	SAPUI5 Fiori App		✓	✗
Currency	defineDisplayOrder	Transaction		✓	✗
FinancialTransaction	processFXSpotForward	Transaction		✓	✗
MarketData	convertCurrencies	Transaction		✓	✗
FinancialTransaction	processFXOTCOOptions	Transaction		✓	✗

Configure Columns

Show similar target mappings

Create Target Mapping

Configure

Create Reference

Delete

Where Used

Original

2:28 PM 4/11/2019

Click on create reference option at the bottom tab of Launchpad designer.

Once you click on create reference you will get the below popup.

Select Catalog

Z]

X-SAP-UI2-ADCAT:Z_123:S4CMD
X-SAP-UI2-ADCAT:Z_123:S4CMD

X-SAP-UI2-ADCAT:Z_DL_BE_APPS:S4...
X-SAP-UI2-ADCAT:Z_DL_BE_APPS:S4CMD

Z_NONGUI_CATALOG
X-SAP-UI2-CATALOGPAGE:Z_NONGUI_CAT...

Z_Test_NonProd_Catalog
X-SAP-UI2-CATALOGPAGE:Z_Test_NonProd...

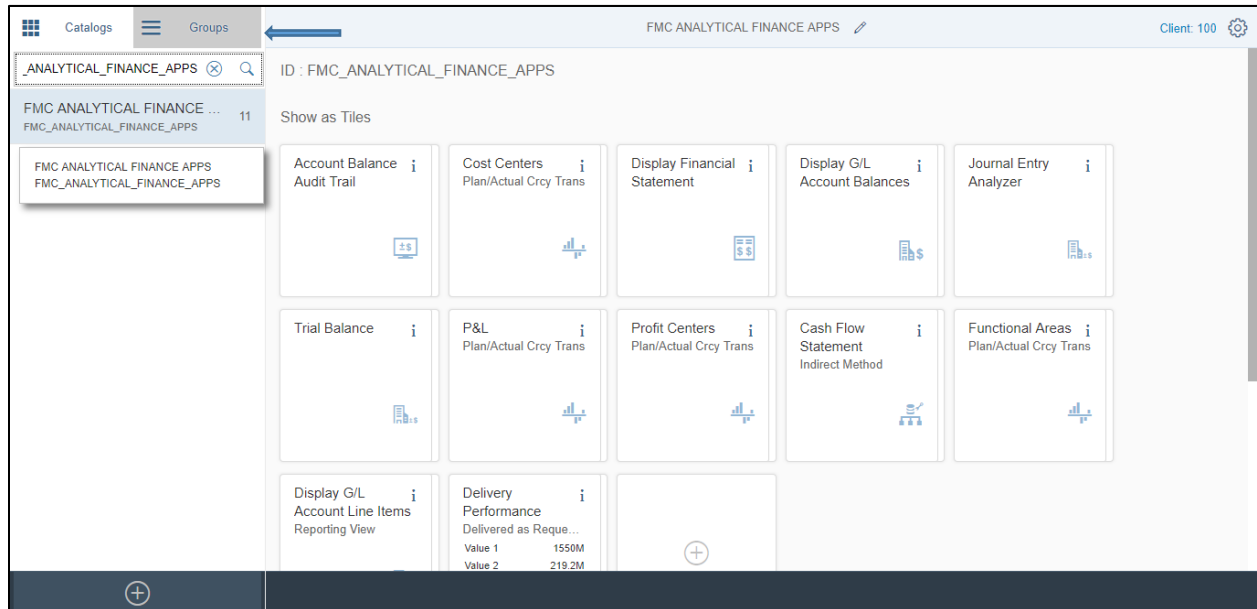
Cancel

Put the Target mapping in the same catalog where we have kept Tile.

For any App to work its tile and target mapping should be in the same catalog.

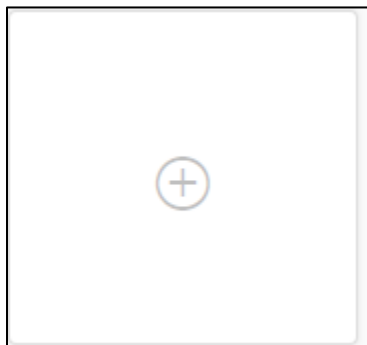
Tiles can offer summarized information about the data available for the users in the app
Depending on the roles of the user certain groups of apps are displayed wherein Target Mapping defines the target application which is launched when an intent (semantic object/action) is triggered

Next, add the tile from this catalog into your group. Go to the Groups section.

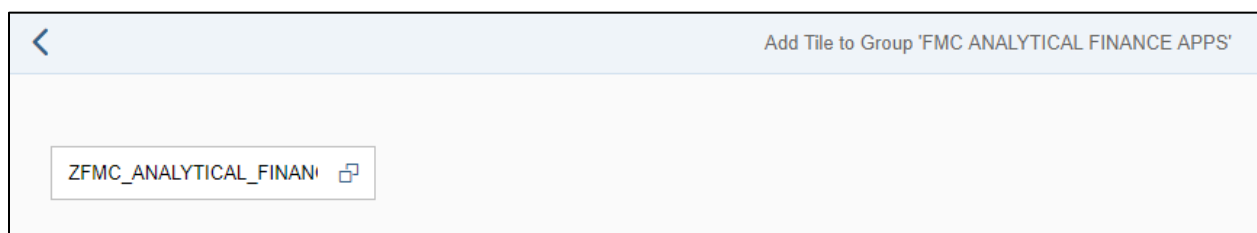


Select the group name in which you want to add the tile.

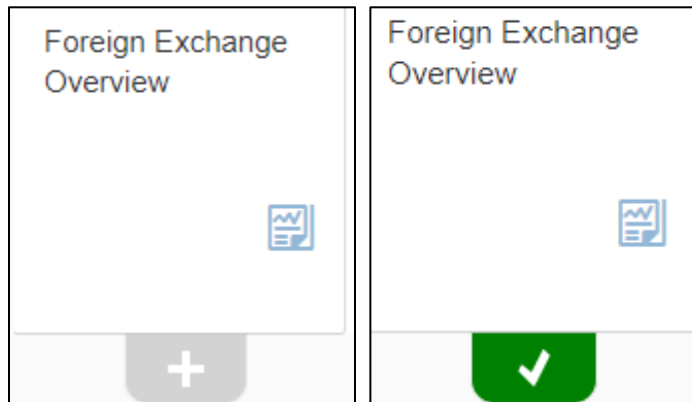
Click on Add Tile



Select the catalog from where you want to pick the tile.



Activate your tile (Just click the + icon to add it in a group)



Group is a subset of catalog that contains the apps visible on the SAP Fiori Launchpad entry page. Which tiles are displayed on a user's entry page depends on the group assigned to the user's role.

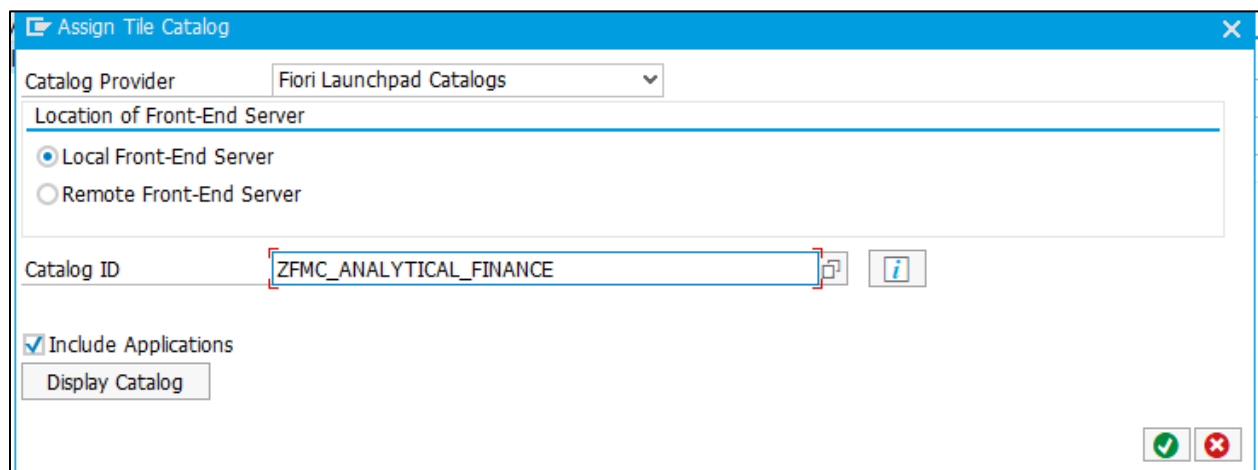
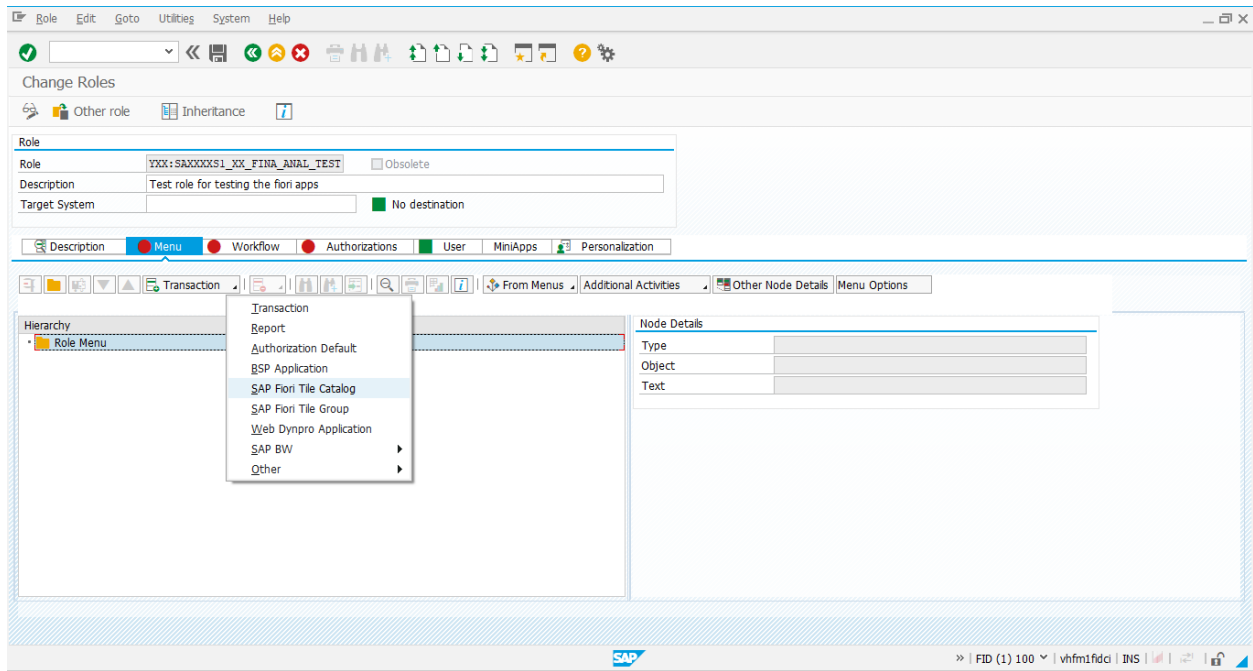
Now the Z-catalog and Z-group is ready and we can provide access to the users via roles.

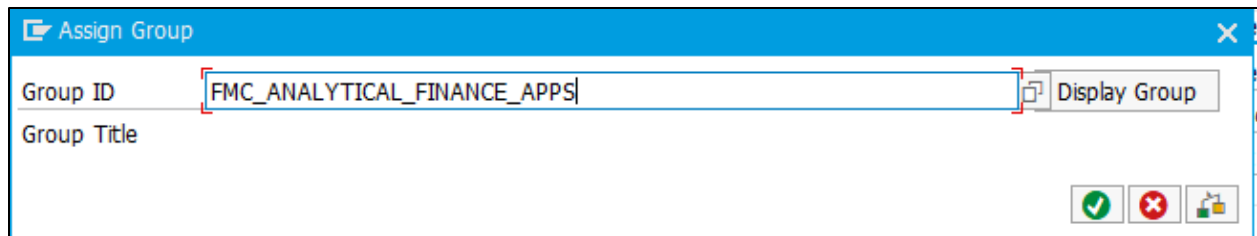
Roles and Authorization Concept

Fiori Role Creation (In Front end)

On the Front-end Fiori Role, add the catalog ID and Group ID from the Menu Tab.

Go to PFCG>Create a Role>Menu>Small Drop Down at Transaction Button>SAP Fiori Catalog





Click on Save.

Assign this Front-end role to the user.

Back-end Configuration (In S4 HANA)

Create one role on the backend (s4 system) and add the following depending on the Fiori application type.

Configuration of backend PFCG role is different for different type of apps.

Below is the demonstration for each cases.

Case 1: Application Type: SAP GUI Based App

Case 2: Application Type: Web Dynpro Application

Case 3: Application Type: Transactional Apps

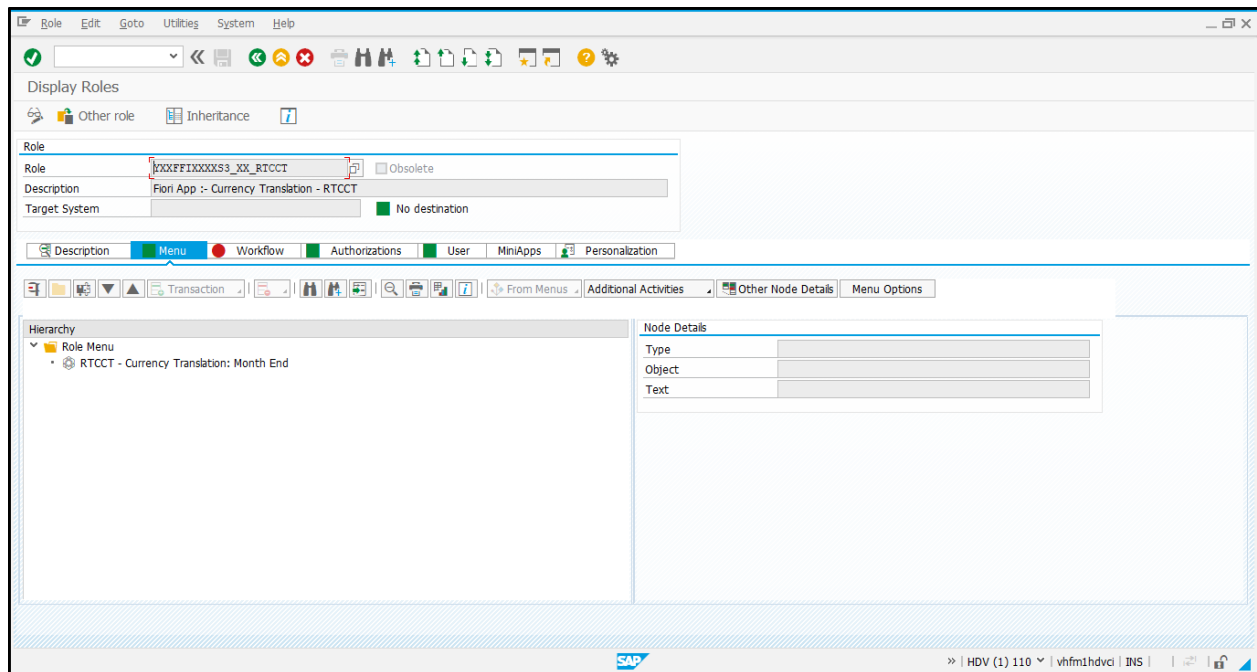
Case 4: Application Type: Analytical Apps (Design Studio Apps with BEX Queries)

Case 5: Application Type: Factsheets

Case 1: Application Type > SAP GUI App

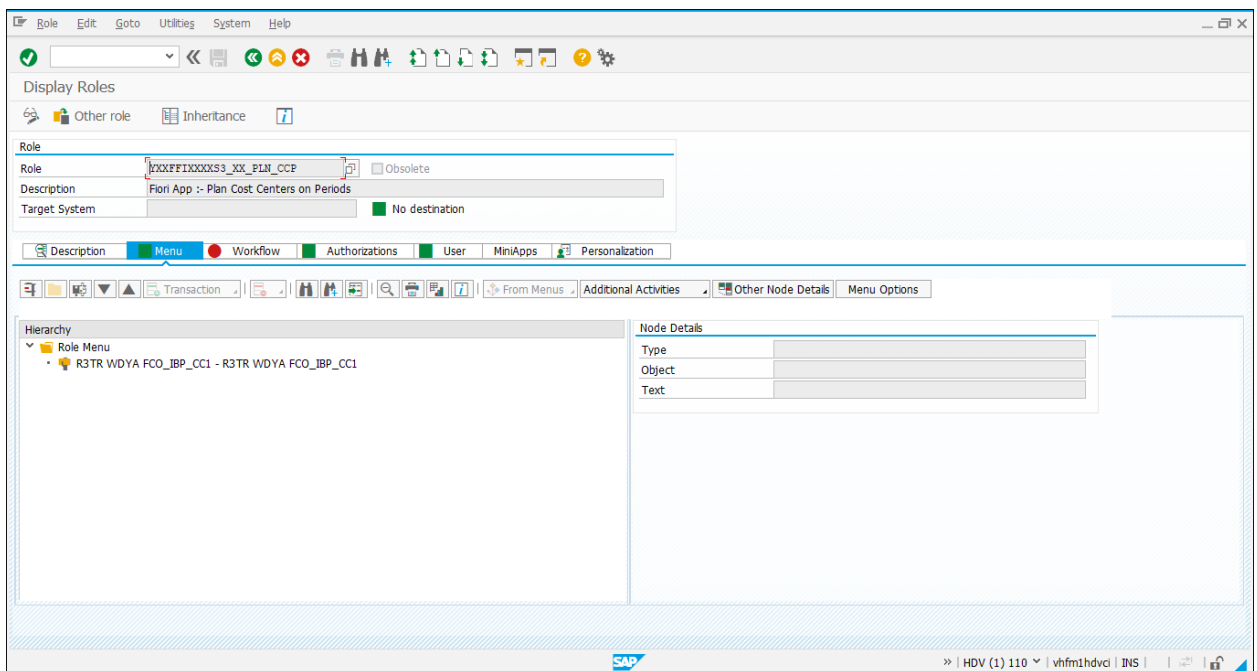
For SAP GUI app only T-code access should be enough at the backend role.

Create a PFCG role and add the transaction code in the Menu Tab. Maintain the authorization data & generate the profile.



Case 2: Application Type > Web Dynpro Application

Go to Menu -> Authorization Default -> Web Dynpro Application

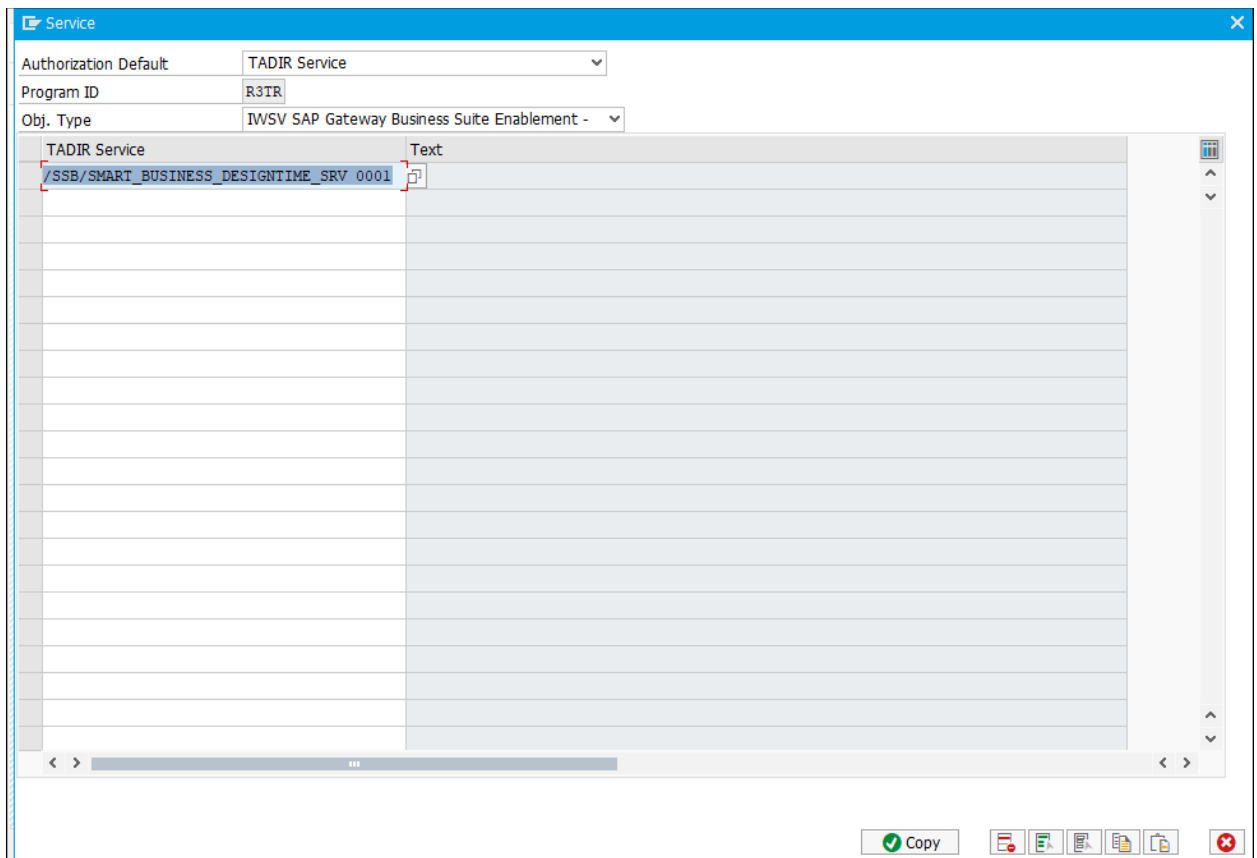
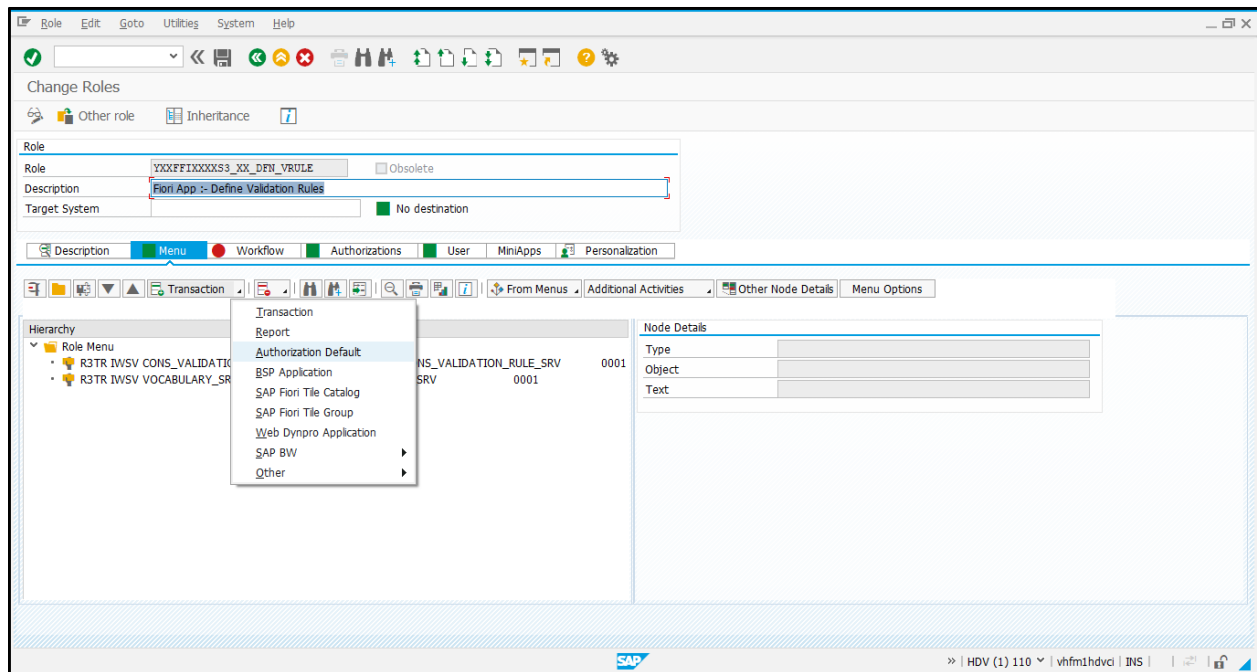


[illegible]

Add the O-data service under TADIR service.

Case 3: Application Type: Transactional Apps (add the O-data services)

Go to Menu -> Authorization Default -> TADIR Service -> Obj type - IWSV



Note: Once you add the O-data Services the relevant object **S_SERVICE** with the exact filed value with respect to the app for belongs to O-data will get pulled out in authorization tab.

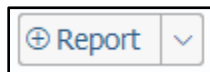
Once done please generate the role and assign the role to user.

Case 4: Application Type > Analytical Apps

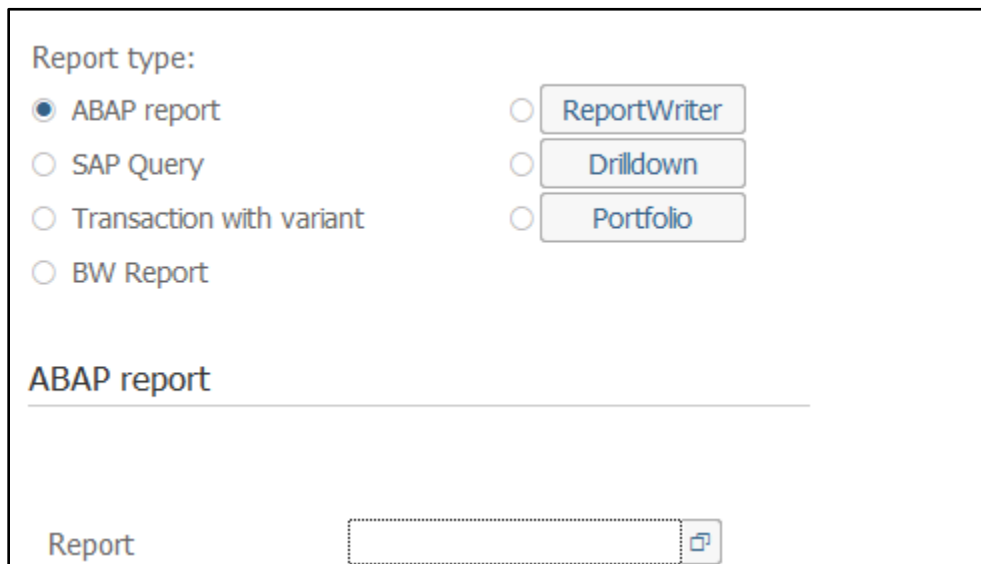
Analytical App are of 2 type once is only analytical apps and one is Analytical Design Studio app.

Analytical app will work only by adding the O-data till same as previous step, But for Design Studio App we would be needing one extra object needs to be add with BEX Queries value.

Menu -> from the transaction tab dropdown select Report ID (Enter the report name).

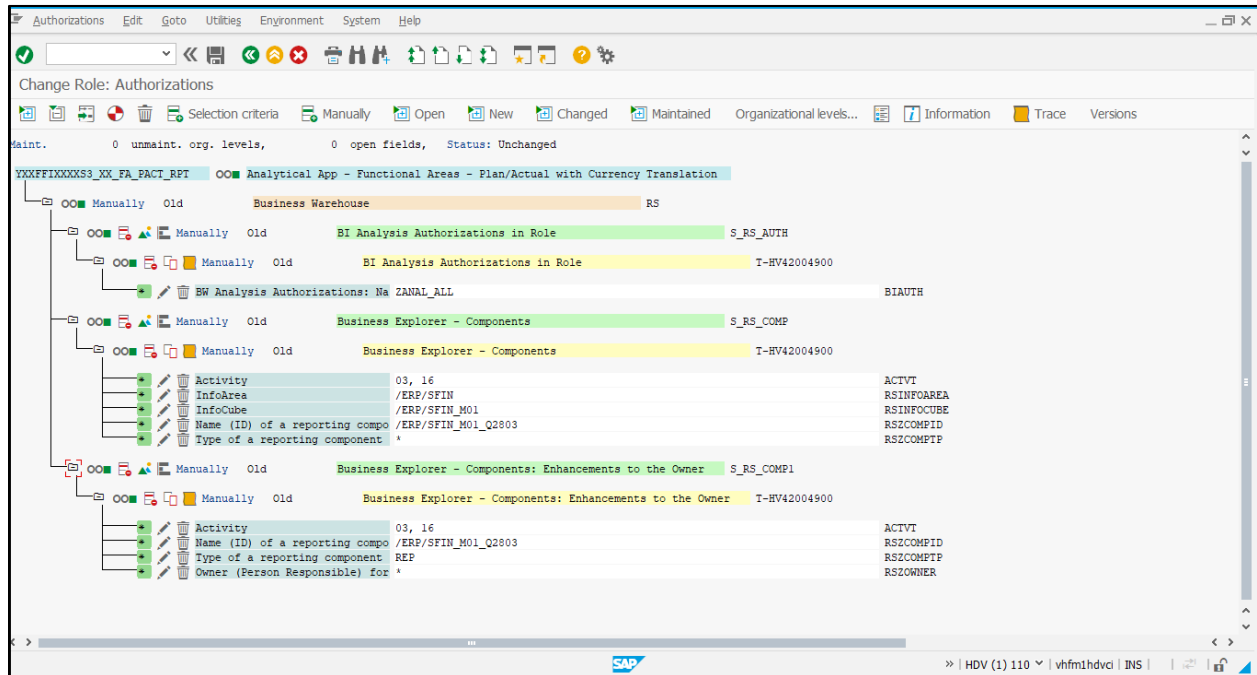
A small rectangular button with a light blue background. It contains the text "Report" in blue, followed by a small downward-pointing chevron icon.

Report ID (Enter the report name).

A dialog box for configuring a report. It has a title "Report type:" and four radio button options: "ABAP report" (selected), "SAP Query", "Transaction with variant", and "BW Report". To the right of these are three buttons: "ReportWriter", "Drilldown", and "Portfolio". Below the radio buttons is a section titled "ABAP report" with a horizontal line underneath. At the bottom left is the label "Report" and a text input field with a dotted border. To the right of the input field is a small icon of a document with a plus sign.

One you add the report the related object will get pulled out in authorization tab.

Maintain the values for S_RS_COMP, S_RS_COMP1 and S_RS_AUTH in the Authorization tab.



Prerequisite for working of Design Studio Apps: BW component must be activated in the system from system configuration stand point.

Case 5: Application Type > Factsheets

Factsheets app won't be available in the form of tile.

In the designer Launchpad inside the Catalogs, there will be only Target Mapping available for Factsheet Apps.

Role creation is similar to the transactional app.

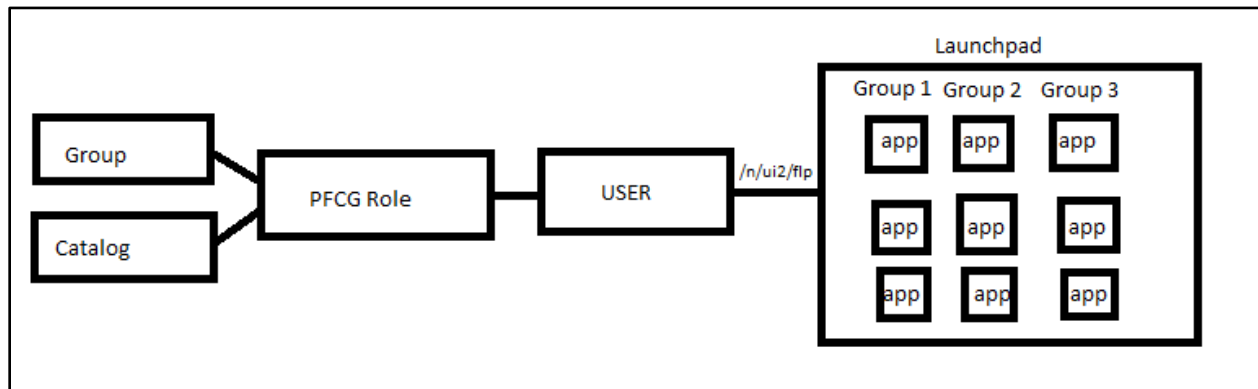
Go to Menu -> Authorization Default -> TADIR Service ->

Maintain the app template in the object S_ESH_CONN in the authorization tab.

User Creation and Role Assignment

Create same user id, both on the Front end and Back end server and assign Front end role on the Front system and backend role on the Backend system to the user along with the Default role.

How the Tiles we have configured in the catalog and group will be visible to the users at the Launchpad/Fiori Home Page of end users, you can see the below diagram.



Trouble Shooting of Fiori Apps

There will be number of different types of issues in Fiori apps you will find across after configuration of Tile and Target mapping.

Let's take one example of common app issue,

App ID F1660 - Purchasing Group Activity

Purchasing Group Activities

for Strategic Buyer

SAP S/4HANA ▾

Required Back-End Product SAP S/4HANA

Line of Business Sourcing and Procurement, Supply Chain

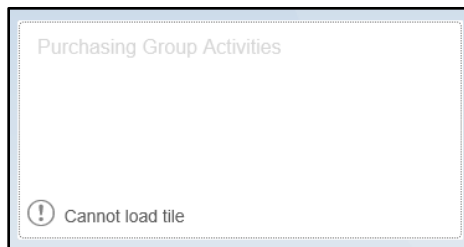
Application Type Analytical (SAP Fiori: SAP Smart Business generic drill down app)

Database HANA DB exclusive

Form Factor Desktop, Tablet

App ID F1660

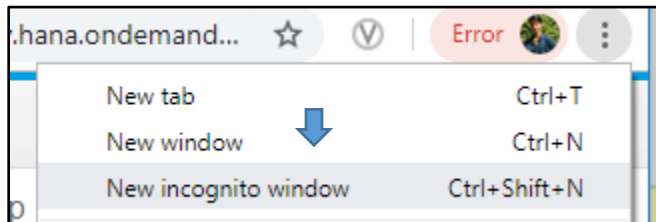
In the Launchpad you can see,



Error Message – “cannot load tile” = Common issue for 80% of the apps.

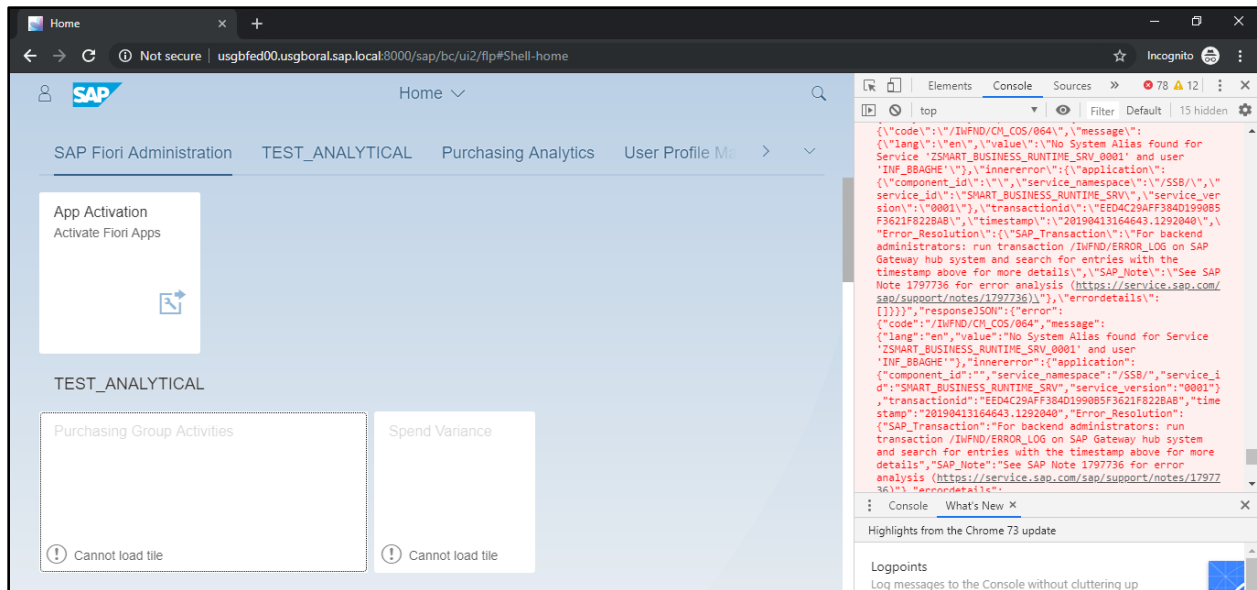
Suggestion before we start troubleshooting,

Always use Incognito Mode in Chrome for troubleshooting as Incognito mode don't save any cache in memory.



Now open the Launchpad and click **F12** button from Key board.

You will get below screen in the left side of the panel you can see console tab, go to Console tab.



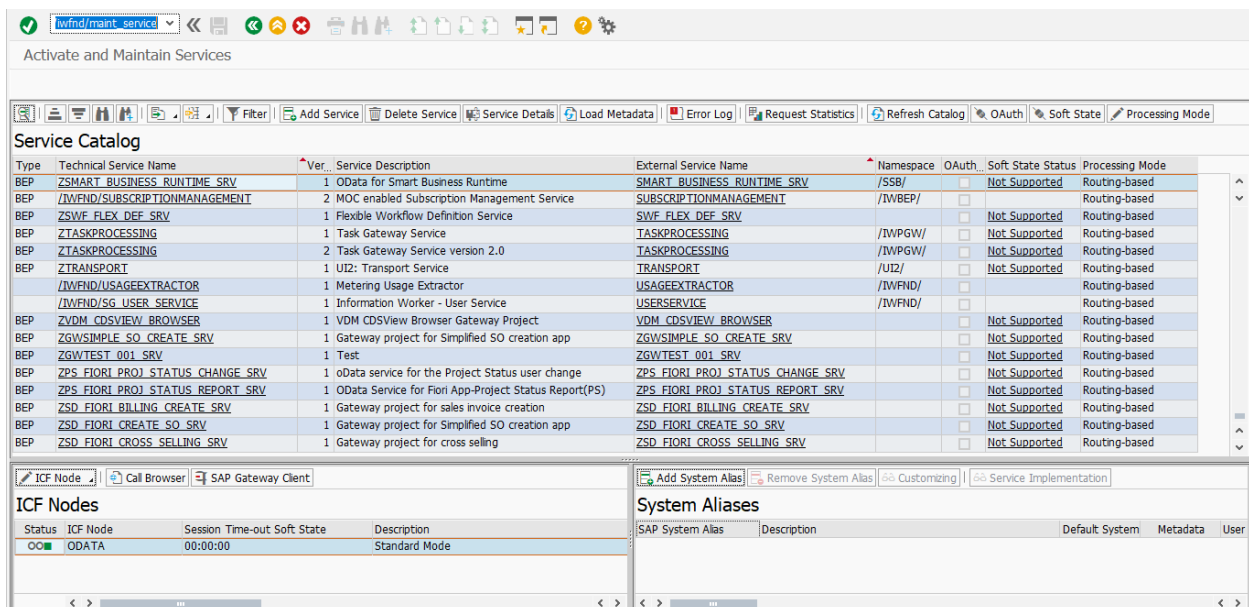
Now Once you click the tile which is giving error "Can't Load Tile"
 You can see the root cause of the error in console tab as below,
 Like in this case,

```
{
  "lang": "en",
  "value": "No System Alias found for Service 'ZSMART_BUSINESS_RUNTIME_SRV_0001' and user 'INF_BBAGHE'",
  "innererror": {
    "application": "ZSMART_BUSINESS_RUNTIME_SRV_0001"
  }
}
```

No System Alias found for service "ZSMART_BUSINESS_RUNTIME_SRV_0001"

FYI - ZSMART_BUSINESS_RUNTIME_SRV is a O-data service.

Go to T-code **/n/iwfnd/maint_service**



You can see for the ZSMART_BUSINESS_RUNTIME_SRV service in the right bottom - No System Alias mentioned.

Now once you maintain the system alias in the O-data

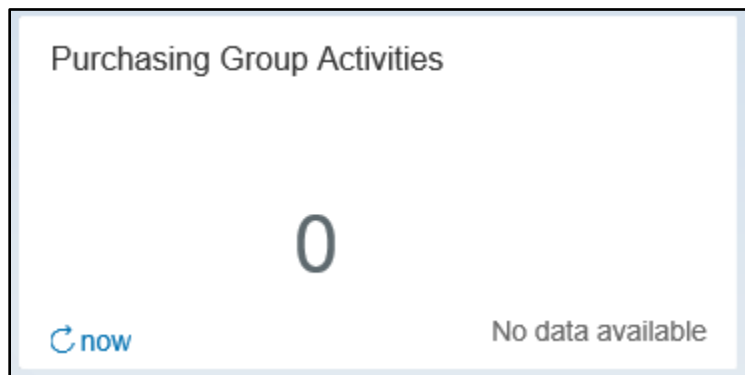
The screenshot shows the SAP Service Catalog and System Aliases configuration. The Service Catalog table lists various services, and the System Aliases table shows the configuration for the ODATA system alias.

Type	Technical Service Name	Ver.	Service Description	External Service Name	Namespace	OAuth	Soft State	Status	Processing Mode
BEP	ZSMART_BUSINESS_RUNTIME_SRV	1	OData for Smart Business Runtime	SMART_BUSINESS_RUNTIME_SRV	/SSB/			Not Supported	Routing-based
BEP	/IWFND/SUBSCRIPTIONMANAGEMENT	2	MOC enabled Subscription Management Service	SUBSCRIPTIONMANAGEMENT	/IWBEP/				Routing-based
BEP	ZSWF_FLEX_DEF_SRV	1	Flexible Workflow Definition Service	SWF_FLEX_DEF_SRV				Not Supported	Routing-based
BEP	ZTASKPROCESSING	1	Task Gateway Service	TASKPROCESSING	/IWPGW/			Not Supported	Routing-based
BEP	ZTASKPROCESSING	2	Task Gateway Service version 2.0	TASKPROCESSING	/IWPGW/			Not Supported	Routing-based
BEP	ZTRANSPORT	1	UI2: Transport Service	TRANSPORT	/UI2/			Not Supported	Routing-based
BEP	/IWFND/USAGEEXTRACTOR	1	Metering Usage Extractor	USAGEEXTRACTOR	/IWFND/				Routing-based
BEP	/IWFND/SG_USER_SERVICE	1	Information Worker - User Service	USERSERVICE	/IWFND/				Routing-based
BEP	ZVDM_CDSVIEW_BROWSER	1	VDM CDSView Browser Gateway Project	VDM_CDSVIEW_BROWSER				Not Supported	Routing-based
BEP	ZGW/SIMPLE_SO_CREATE_SRV	1	Gateway project for Simplified SO creation app	ZGW/SIMPLE_SO_CREATE_SRV				Not Supported	Routing-based
BEP	ZGWTEST_001_SRV	1	Test	ZGWTEST_001_SRV				Not Supported	Routing-based
BEP	ZPS_FIORI_PROJ_STATUS_CHANGE_SRV	1	oData service for the Project Status user change	ZPS_FIORI_PROJ_STATUS_CHANGE_SRV				Not Supported	Routing-based
BEP	ZPS_FIORI_PROJ_STATUS_REPORT_SRV	1	OData Service for Fiori App-Project Status Report(PS)	ZPS_FIORI_PROJ_STATUS_REPORT_SRV				Not Supported	Routing-based
BEP	ZSD_FIORI BILLING_CREATE_SRV	1	Gateway project for sales invoice creation	ZSD_FIORI BILLING_CREATE_SRV				Not Supported	Routing-based
BEP	ZSD_FIORI_CREATE_SO_SRV	1	Gateway project for Simplified SO creation app	ZSD_FIORI_CREATE_SO_SRV				Not Supported	Routing-based
BEP	ZSD_FIORI CROSS SELLING_SRV	1	Gateway project for cross selling	ZSD_FIORI CROSS SELLING_SRV				Not Supported	Routing-based

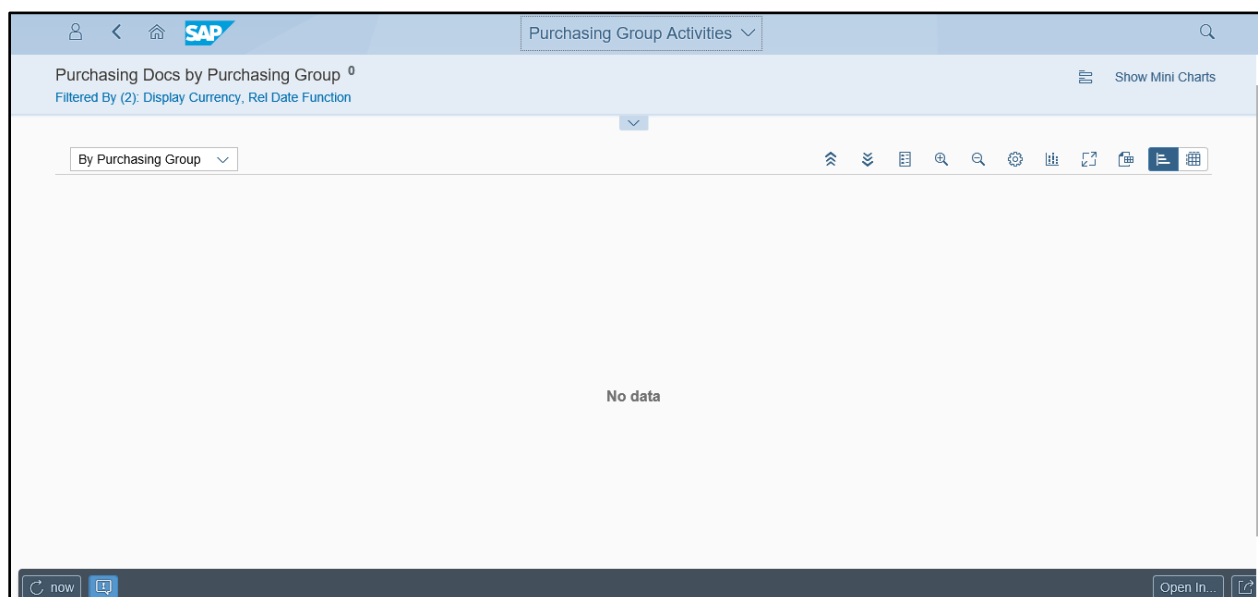
Status	ICF Node	Session Time-out	Soft State	Description
OO	ODATA	00:00:00		Standard Mode

SAP System Alias	Description	Default System	Metadata	User
S4FIN	S4FIN	✓	✓	

You will now be able to see the tile is getting loaded successfully.



And Its opening successfully.



No Data is showing because this is a sandbox system where in no data has been configured from functional side.

As far as we have seen the **F12** button is a very use full for inspecting the errors.

There is one another way of finding the cause of error,

Go to T-code **/n/iwfnd/error_log**

For the same above case

The screenshot displays the SAP Gateway: Error Log interface. The top section shows a list of errors with columns for Line, Entry, Date, Time, User, T100 Error ID, T100, Error C., ICF No., HTTP, and Error Text. The bottom section shows the Error Context for a selected error, including fields like Exp. Name, Value, and a link to SAP Note 1797736.

Line	Entry	Date	Time	User	T100 Error ID	T100	Error C.	ICF No.	HTTP	Error Text
10	1	04/14/2019	00:46:43	INF_BBAGHE	/IWFND/CM_COS064	?	27	odata	500	No System Alias found for Service 'ZSMART_BUSINESS_RUNTIME_SRV_0001' and user 'INF_BBAGHE'
9	1		00:46:42	INF_BBAGHE	/IWFND/CM_COS064	?	27	odata	500	No System Alias found for Service 'ZSMART_BUSINESS_RUNTIME_SRV_0001' and user 'INF_BBAGHE'
8	1			INF_BBAGHE	/IWFND/CM_COS064	?	27	odata	500	No System Alias found for Service 'ZSMART_BUSINESS_RUNTIME_SRV_0001' and user 'INF_BBAGHE'
7	1			INF_BBAGHE	/IWFND/CM_COS064	?	27	odata	500	No System Alias found for Service 'ZSMART_BUSINESS_RUNTIME_SRV_0001' and user 'INF_BBAGHE'
6	1			INF_BBAGHE	/IWFND/CM_COS064	?	27	odata	500	No System Alias found for Service 'ZSMART_BUSINESS_RUNTIME_SRV_0001' and user 'INF_BBAGHE'
5	1			INF_BBAGHE	/IWFND/CM_COS064	?	27	odata	500	No System Alias found for Service 'ZSMART_BUSINESS_RUNTIME_SRV_0001' and user 'INF_BBAGHE'
4	1		00:46:41	INF_BBAGHE	/IWFND/CM_COS064	?	27	odata	500	No System Alias found for Service 'ZSMART_BUSINESS_RUNTIME_SRV_0001' and user 'INF_BBAGHE'
3	1			INF_BBAGHE	/IWFND/CM_COS064	?	27	odata	500	No System Alias found for Service 'ZSMART_BUSINESS_RUNTIME_SRV_0001' and user 'INF_BBAGHE'
2	1			INF_BBAGHE	/IWFND/CM_COS064	?	27	odata	500	No System Alias found for Service 'ZSMART_BUSINESS_RUNTIME_SRV_0001' and user 'INF_BBAGHE'

Exp. Name	Value
..ERROR_CONTEXT	
..ERROR_INFO	RFC Error: Illegal destination type 'H'.
..ERROR_RESOLUTION	
...SAP_NOTE	See SAP Note 1797736 for error analysis
...LINK_TO_SAP_NOTE	https://service.sap.com/sap/support/notes/1797736
..Exceptions	
.../IWFND/CM_MGW_TECH_EXCEPTION	
....Text	RFC Error: Illegal destination type 'H'.
....Attributes	
.....GC_STATUS_NOT_IMPLEMENTED	501
.....STATUS_CODE_UNAVAILABLE	500

After reproducing the error in Launchpad you can see the error log in gateway system.

Same error message – No system alias found for O-data service.

Different type of Business Requirements in Fiori

1.

- Apart for configuring the tiles and app in Fiori.
- Users has the ability to customize their own home page by rearranging the tiles in a group.
- Users can make their own favorites list.

But this is a drawback in any business scenario during the training due to the lack of knowledge end users always mess/delete the arrangement of the tiles and they always come up to the security team for the resolution.

So better approach to restrict the modification of Fiori home page during the training time.

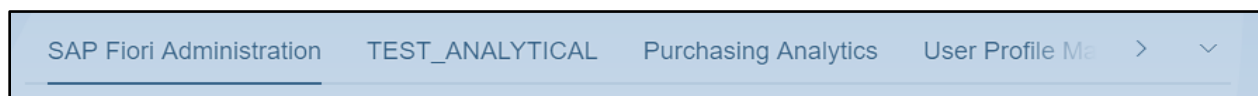
To disable the personalization of users please assign below Target mapping.

/UI2/CONFIG_PERS_OFF

2.

Some users will be having the 20 groups or n number of groups in their homepage based on their business role and mapping of the role,

But in the first screen they can only see 4-5 groups at a time.



To explore more groups, they need to scroll down the > sign.

Some of the Tiles and Groups are supposed to be the first in the sequence as per the business process.

Here comes the sequencing requirement from the business.

So the question is how we can decide in the Launchpad which group will appear first or in a defined sequence?

To answer the above question we need to set the priority of groups sequence inside the front end PFCG role.

Below is the example of Trade Store Operator Role,

Change Roles

Other role Inheritance

Role

Role: Z_GRP_TS_OPERATOR ☐ Obsolete

Description: Trade Store Operator-Fiori Group

Target System: ☐ No destination

Description Menu Workflow Authorizations User MiniApps Personalization

Transaction From Menus Additional Activities Other Node Details Menu Options

Hierarchy

- Role Menu
 - Sales/UBLSalesEnquiryExecute - Sales/UBL Sales Enquiry Execute
 - Sales/UBLSalesOrderExecute - Sales/UBL Sales Order Execute
 - SalesDeliveryExecute - Sales Delivery Execute
 - SalesShipmentCostExecute - Sales Shipment Cost Execute
 - SalesBillingGeneral - Sales Billing General
 - SalesMasterGeneral - Sales Master General
 - MaterialMstrTmsprtPricingGnrl - Material Master Transport Pricing General
 - MaterialMasterGeneral - Material Master General
 - SalesSalesCreditExecute - Sales Sales Credit Execute
 - SupplyChainInvntryCntrlExec - Supply Chain Inventory Control execute
 - ProcurementPOGeneral - Procurement PO General
 - SupplyMasterGeneral - Supply Master General
 - SupplyChainInvntryCntrlGeneral - Supply Chain Inventory Control General
 - SupplyChainStocktakeExecute - Supply Chain Stocktake Execute
 - StockTransportPOExecute - Stock Transport PO Execute

Node Details

Type	
Object	
Text	

User the role menu all the groups in assigned in the sequence based on the business inputs. Same as above sequence the groups will appear in the Launchpad see below.

Home

SAP Home

COMMON Sales/UBL Sales Enquiry Execute Sales/UBL Sales Order Execute Sales Delivery Execute Sales Shipment Cost Execute

Create Inquiry Change Inquiry Display Inquiry List Inquiries

- In the same way Sequencing of Tiles inside the group can also be done. Just an example of Sales Enquiry Execute group.

Sales/UBL Sales Enquiry Execute Client: 100

ID : Sales/UBLSalesEnquiryExecute

Show as Tiles

Create Inquiry Change Inquiry Display Inquiry List Inquiries

+

You can see in the previous screenshot the sequence we have maintained inside the group is coming in the Launchpad/Homepage of the users.

Some of the Important T-code in Fiori

/UI2/CACHE_DEL	Delete cache entries
/UI2/CUST	Customizing of UI Technologies
/UI2/FLC	Fiori Launchpad Checks
/UI2/FLIA	Fiori Launchpad Intent Analysis
/UI2/FLP	SAP Fiori Launchpad
/UI2/FLP_CONTCHECK	Fiori Launchpad – Content Checks
/UI2/FLP_INTENTCHECK	Fiori Launchpad – Intent Checks
/UI2/FLPD_CONF	Fiori Launchpad Designer (cross-client)
/UI2/FLPD_CUST	Fiori Launchpad Designer (client-specific)
/UI2/GW_ACTIVATE	Gateway – Activation
/UI2/GW_APPS_LOG	Gateway – Application Log
/UI2/GW_ERR_LOG	Gateway – Error Log
/UI2/GW_MAINT_SRV	Gateway – Service Maintenance
/UI2/GW_SYS_ALIAS	Gateway – Manage SAP System Alias
/UI2/NWBC	Start UI2 NWBC
/UI2/SEMOBJ	Define Semantic Object – Customer

SE16

/IWFND/V_MGDEAM - O-data vs System alias

/IWFND/V_DFSYAL -- Display View Manage SAP System Aliases

USR21 – mapping

sap-ui-tech-hint=GUI – App parameter

/UI2/V_ALIASCAT

/UI2/VC_ALIASCAT

/UI2/V_ALIASMAP - for replication

/ui2/fsac - to check rfc

/UI2/GET_APP_DESCR_REMOTE_ALL

SUI_TM_MM_APP & AGR_HIER in SE16

/n/ui2/flpcm – catalog to app

IMP Roles navigation role

SAP_BR_INTERNAL_SALES_REP

SAP_BR_MANAGER

SAP_BR_MANAGER_COST

SAP_BR_RETURNS_REFUND_CLERK

SAP_BR_GL_ACCOUNTANT

SAP_BR_OVERHEAD_ACCOUNTANT

SAP_BR_SALES_ACCOUNTANT

Issues.1

Navigation to this application is not supported

Get error "Navigation to this application is not supported"

Cause

The navigation intent is not present because the backend catalog has not replicated. The alias map does not appear in tCode /nSM30 in the table [/UI2/V_ALIASMAP](#)

Resolution

Maintain the aliases for the S4Fin in the /UI2/V_ALIASMAP table.

Assign SAP standard role, If one app is not getting navigated.

Issue.2

Order to cash Performance Analytical tiles.

No Data issue

Add S_ESH_ADM , POC_AUTH and /UIF/LREP , /UIF/FLEX object in front end and backend in addition to sap standard role.

Replication

/UI2/GET_APP_DESCR_REMOTE

SE38 - PRGN_CREATE_FIORI_FRONTENDROLE – Create Front End roles for Fiori Launchpad

/n/UI2/FLPCM – For Catalog and Group Creation

Execute SGOS, Create new entry for service PCATT_CREA

Under this maintain the value 02 'invisible' and '01' for inactive.

RSBCS_USER_DELETE_JOB - Delete BCS users

Table to check SOUD or SOUC

/IWFND/R_METERING_VIEW – Fiori Usage (<https://blogs.sap.com/2020/07/22/how-to-check-sap-fiori-usage/>)

/UI2/FLP_CONT_MGR – Content manager similar to /ui2/flpcm_cust